

Visualizing on the Wrist: Impact of Motion, Dial Shape and Visualization Type on Smartwatch

Zhouxuan Xia

Fengyuan Liao

Jinyuan Du

Yu Liu^{*†}

Xi'an Jiaotong-Liverpool University, Suzhou, China

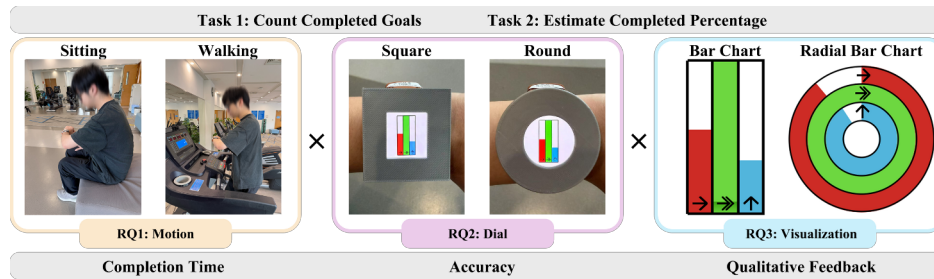


Figure 1: Overview of Experimental Setup and Research Questions (RQs). We studied how Motion (Sitting vs. Walking), Dial (Square vs. Round), and Visualization (Bar vs. Radial Bar) influence performance and experience on a smartwatch.

ABSTRACT

Smartwatches increasingly serve as daily data analysis tools, yet their visualization designs face unique challenges: small screens, diverse dial shapes (round/square), and mobile usage contexts (sitting/walking). These factors may interact in unexpected ways, potentially compromising data readability during real-world use. Through a controlled experiment (N=32) comparing bar vs. radial visualizations across dial shapes and motion states, we found: (1) walking significantly reduces estimation accuracy, (2) bar charts consistently outperform radial variants by better supporting linear perception, and (3) radial bar graphs on square dials create particularly poor affordances than other combination. Our work provides the empirical evidence for how smartwatch dial factors, usage contexts and visualization design jointly impact data interpretation, offering concrete design guidelines for smartwatch visualization. Supplemental material is available at <https://osf.io/k7fa3/>.

Index Terms: Smartwatch Visualization, Mobile Visualization, Human-Computer Interaction

1 INTRODUCTION

Smartwatches have been the most common wearable devices, empowering users to swiftly track a variety of their health and activity metrics with built-in sensors, such as steps, heart rate [14], sleep patterns [16, 17] and calorie expenditure, for timely analysis and decision-making [1, 7, 11]. Their wearable and portable nature distinguishes them from traditional PCs, which are typically confined to stable home or office settings, allowing smartwatches to be used across various times and places [13]. Previous studies have explored the impact of different motion trajectories [15] and speeds [15, 25] on the readability of visualizations on mobile devices, revealing that motion significantly affects readability, particularly in dynamic scenarios such as walking, in more detail, walking negatively impacts reading performance, with reading accuracy significantly declining at higher speeds (e.g., 6km/h). Additionally, a survey [27] showed that round and square smartwatch dials are equally

prevalent in the current market, with linear bar and radial bar charts frequently employed to present activity data. Yet, the interplay between these two visualization types and the two smartwatch dial designs remains unexplored. Thus, this study aims to examine the effects of different smartwatch dial shapes and visualization types on reading activity data in two representative smartwatch usage scenarios: sitting and walking. Our study aim to address the following three key **Research Questions (RQs)** related to smartwatch visualizations (see in Figure 1):

RQ1: What is the effect of *motion* on the users' interpretation of activity data visualizations on smartwatches?

RQ2: What is the effect of smartwatch *dial shape* on the users' interpretation of activity data visualization?

RQ3: What is the effect of *visualization type* on the users' interpretation of activity data on smartwatches?

To explore these RQs, we designed a controlled task-based experiment that examines how users interpret smartwatch activity data under varying conditions. Our study offers two key contributions to the understanding of smartwatch data visualization. Firstly, we extend prior work by systematically examining the combined effects of motion, display shape, and visualization type on smartwatch data interpretation. Secondly, we offer design considerations to support more effective and efficient smartwatch visualization design.

2 RELATED WORK

A wide range of studies has explored how motion or dial shape or visualization type affect interaction and data readability on varied devices or screens. Table 1 in Appendix summarizes key findings.

Effects of Motion. Since the usage of smartwatches and smartphones is not restricted to sitting, many previous studies have explored how motion impacts reading performance. For instance, Mustonen et al. [18] found that text legibility on smartphones decreases during walking. Schildbach et al.'s [20] compared target acquisition and reading performance on smartphones while standing and walking, suggesting a significant decline in task completion speed and accuracy during motion. Crowley's work [9] further explored dual-task interference on mobile phone's usage during walking, highlighting challenges of divided attention in motion contexts. Turner et al. [25] examined text input on smartwatches and reported reduced accuracy when walking. More recently, a study very similar to ours, Grioui et al. [15] conducted controlled experiments on smartwatches, assessing micro-visualization reading performance across sitting, standing, and walking at varying speeds, revealing

^{*}Corresponding Author

[†]Manuscript received xx xx xxxx; revised xx xx xxxx; accepted xx xx xxxx. Date of publication xx xx xxxx.

that higher walking speeds correlate with greater impairments in accuracy and increase speed. Overall, these studies underscore that motion decrease data readability.

Impact of Screen Shape. With the advancement of digital displays, screen shapes have diversified beyond the traditional rectangle. Serrano et al. [21, 22] examined text recognition performance and user preferences for graphical layouts on non-rectangular displays, showing that round screens can increase reading difficulty due to edge curvature. They provided guidelines for improving text legibility and organizing graphical elements effectively on small, uniquely shaped displays. Simon et al. [23] explored information retrieval tasks on non-rectangular interfaces, observing that shapes like circles and triangles impact task efficiency, especially in search tasks, with round interfaces requiring more time. Although these studies focused on larger devices like laptops, they provide a theoretical foundation for comparing square and round smartwatch faces.

Radial Versus Linear Visualizations. Previous studies had conducted many comparative evaluations of radial and linear visualizations on different devices. Blascheck et al. [2, 4] investigated glanceable visualizations on smartwatch and laptop, finding that bar charts were faster for rapid data comparisons and subjectively preferred bar charts [16]. Their later work [3] expanded on part-to-whole glanceable visualizations, confirming bar charts' strengths and offering detailed design insights. Also, Niess's work [19] demonstrated that bar charts are more effective than radial charts in promoting reflection, supporting their use in smartwatches. Additionally, available research [26, 6, 5] compared radial and linear charts, consistently finding that linear formats (e.g., bar charts) outperform radial ones in task accuracy and speed when considering task type and cognitive load, though radial charts are more compact.

Gaps. Prior studies have examined motion's effect on reading performance but overlooked how dial shape and visualization type, or their interplay with motion, influence smartwatch data interpretation. Our study addresses this by systematically exploring these combined factors.

3 USER STUDY

This study examines how motion, dial shape, and visualization type influence users' performance and experience when reading activity overview data on a smartwatch. We chose two representative tasks on the activity overview interface from a prior survey [27]: counting completed activity goals and estimating the completed percentage of goal. The experiment followed a $2 \times 2 \times 2$ within-subjects design with three independent variables: motion (sitting  vs. walking at a constant 3km/h  on a treadmill), dial shape (square  vs. round  smartwatch dials) and visualization type (bar chart  vs. radial bar chart ). The walking speed was selected based on average natural walking paces measured across 5 participants with varying heights and weights, and was informed by Grioui et al. [15]. The study was conducted in a school gymnasium. To minimize environmental interference, we positioned the treadmill away from windows to reduce lighting variability. To mitigate learning and order effects, we counterbalanced the presentation order of task type, visualization type, dial shape, and motion condition using a Latin square design. This ensured a systematic variation of trial sequences across participants. The study protocol was reviewed and approved by the university's ethics committee.

PROCEDURE. The experiment began with an introduction by the study coordinators, who explained the purpose and overall structure of the study. Once participants confirmed their willingness to participate, they signed a consent form and completed a demographic questionnaire collecting information on gender, age, color vision, academic background, and smartwatch usage habits. Following this, participants entered a tutorial phase in which they were introduced to experimental tasks: Counting completed goals and

estimating completed percentage. Each participant completed 10 practice trials for each task to familiarize themselves with the task format. Practice trials were structurally identical across all participants and included opportunities for clarification. Each trial followed a three-step structure (see Figure 4 in Appendix): (1) presentation of a short task description; (2) display the corresponding visualization stimuli; and (3) the participant's response via voice input. In Task 1, participants verbally reported a number between 0 and 3 to indicate the number of completed goals. In Task 2, they gave a spoken percentage estimate based on the visualization. All voice responses were recorded by a recording pen.

During the practice phase, participants were encouraged to ask questions, which were answered by the coordinators. However, during the formal trials, the coordinators observed silently to avoid influencing participants' performance. Participants had short rest breaks of approximately 5 minutes in the middle to reduce fatigue and support sustained performance. At the conclusion of the experimental trials, participants took part in a brief semi-structured interview to provide qualitative feedback. Questions focused on how motion, dial shape, and visualization type influenced their interpretation strategies. Each participant spent roughly 55 minutes in the study.

TASK. We designed two tasks to reflect typical smartwatch activity overview interface usage based on previous research on activity overview interfaces [27]. In both tasks, participants responded via voice input. Tasks were performed under all combinations of motion, dial shape and visualization type. They were instructed to consider both speed and accuracy, no corrective feedback was provided after each response.

Task 1: Count Completed Goals. In this task, participants identified how many activity goals had been completed by interpreting the visualizations shown on the smartwatch. A goal was considered completed when its visual progress indicator appeared fully filled.

Task 2: Estimate Completed Percentage. In this task, participants estimated the completed percentage of a specific activity item, visually highlighted by a black dot, following prior work [2, 24, 8].

STIMULI GENERATION. For *Task 1 (Counting Goal)*, we generated 54 stimuli based on two visualization types (Bar Chart and Radial Bar Chart). Each visualization displayed three distinct data items representing different levels of goal completion. Values were drawn from three magnitude ranges commonly found in smartwatch activity data (see Figure 5 in Appendix): LOW (0%–50%), HIGH (51%–100%), and EXTRA (> 100%, capturing overachievement). All possible range combinations ($3 \times 3 \times 3$) were systematically generated for each visualization type, resulting in 27 combinations per type, and 54 unique stimuli in total. For *Task 2 (Estimating Value)*, we introduced a target marker (a black dot) to highlight one of the three data items in each visualization, requiring participants to estimate the indicated value. Therefore, the Task 2 stimulus was extended three times by randomly assigning the marker to one of the three data positions, yielding 162 stimuli (54×3). All stimuli maintained consistent color schemes, which referred to the Apple Watch design, and dimensions, and were presented randomly to minimize sequence effects.

PARTICIPANTS. We recruited 32 participants (13 males and 19 females); ranged 19–26 ($M=21.38$) with diverse background from a local university through campus flyers and social media, detailed information is provided in Table 2 and Table 3 in Appendix. Participants were selected based on their self-reported experience with wearable devices and physical exercise, aiming to reflect the typical smartwatch user demographic. Eligibility criteria included self-reported normal or corrected-to-normal vision and regular use of a smartwatch or regular engagement in physical exercise. Thirty-one participants owned a smartwatch, with shapes being rectangular (12), circular (11), and square (8). Most participants reported wearing the smartwatch on their left arm, with only one participant

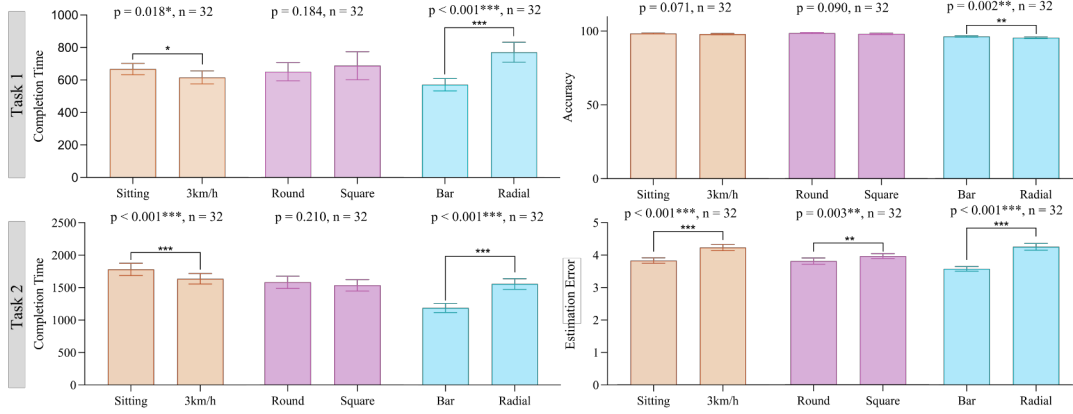


Figure 2: Top: Mean Completion Time (ms) and Accuracy (%) for Task 1. Bottom: Mean Completion Time (ms) and Estimation Error (%) for Task 2. Error bars represent 95% Bootstrap confidence intervals (CIs) with each p-value indicated.

wearing it on the right. During the experiment, participants wore the smartwatch on the same arm as indicated in their questionnaire responses to maintain ecological validity.

APPARATUS. The study was conducted using an OPPO Watch X running Android Wear 8.1.0. Participants performed the walking condition on a treadmill set to a constant speed of 3 km/h, supervised by researchers for safety. To simulate different dial designs while controlling for display size, we fabricated two detachable gray dials using 3D printing (see Figure 6 in Appendix): one square and one round. Both dials were designed to ensure the following consistency across conditions: The visualization content occupied the same screen area ($\approx 264\text{mm}^2$) within both dial types. The viewable screen area for visualizations (the display window cutout) was identical ($\approx 648\text{mm}^2$) between the two dials. The overall device footprint (including the printed outer frames) ($\approx 2750\text{mm}^2$) were matched to maintain comparable device size perception.

METRICS. We collected and analyzed a combination of quantitative and qualitative metrics throughout the study.

Completion Time: The time taken to complete each trial was recorded in milliseconds (ms), starting from the appearance of the visualization (View 2 in Figure 4) until the participant proceeded to View 3, prior to giving their verbal response. This was designed to capture the core perceptual and decision-making phase, excluding variability due to speech. We adopted this approach because once participants confirmed and advanced to the input view, they could not return, ensuring that the recorded time accurately reflected their interpretation of the visualization. All timings were automatically logged using a built-in timer in our Java program.

Accuracy: For each task, we computed accuracy based on the correctness of participants' verbal responses. In Task 1, accuracy was defined as whether the participant correctly reported the number of completed goals and calculated the average accuracy of whole trials (0 = false, 1 = true). For Task 2, we defined accuracy as estimation error, which was measured by the absolute difference between the estimated (%) and actual value (%) of the highlighted item.

Qualitative Feedback through Interviews: After completing the experiment, participants took part in brief semi-structured interviews, which were recorded and transcribed. Two researchers then open-coded the responses to identify common reading strategies, preferences, and subjective impressions, providing contextual depth that complements the quantitative findings.

HYPOTHESIS. Building upon prior research on smartwatch interaction and visualization design [15, 3, 12, 10], linear formats like bar charts have often been associated with higher performance in comparison tasks [2, 26]. Motion can affect task performance, with accuracy generally declining, and completion time showing a non-linear pattern—often increasing at lower speeds but decreasing at higher speeds [15, 20]. Non-rectangular displays often perform

poorly [21, 23], the specific impact of dial shape on smartwatch data visualization remains largely underexplored. We formulated the following hypotheses. **H1 (Effect of Motion):** Walking will reduce task completion time due to increased urgency and focus (H1a), but decrease accuracy due to limited attention and movement interference (H1b). **H2 (Effect of Dial):** Square dials will outperform round dials. **H3 (Effect of Visualization):** Bar charts will lead to shorter completion time and yield higher accuracy than radial charts. **H4 (Dial-Visualization Fit):** Bar charts will perform better on square dials and radial bar charts will perform better on round dials. **H5 (Combination):** Radial bar charts on round dials will lead to better performance than other combinations while walking.

4 RESULT

All statistical analyses were conducted using IBM SPSS (version 29.0.1.0). Data normality was assessed using the Shapiro–Wilk test, which indicated non-normal distributions across all conditions. Accordingly, non-parametric Wilcoxon Signed-Rank tests were employed for within-subject comparisons. Bonferroni corrections were applied to adjust for multiple comparisons, controlling the family-wise error rate. For descriptive reporting, mean values and 95% confidence intervals (CIs) were computed using 1000 bootstrap resamples. Statistical significance thresholds are denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Results are presented in Figure 2, with Table 4 and Table 5 provided detailed data in Appendix.

Task 1: Count Completed Goals. **Completion Time:** Completion time significantly differed across many conditions. Participants completed the task faster while walking (M = 616.03ms [572.58, 652.46]) compared to sitting (M = 667.33ms [634.02, 700.12]), with a p-value of .018*. No significant difference was observed ($p = .184$) between round (M = 651.25ms [593.38, 703.91]) and square dial (M = 688.40ms [611.12, 774.23]). Additionally, a significant difference was found between visualizations ($p < .001***$), with Task 1 using bar charts (M = 575.04ms [537.79, 613.59]) completed faster than those using radial bar charts (M = 776.16ms [713.01, 834.39]). **Accuracy:** Accuracy did not significantly differ between the sitting (M = 98.35% [98.02, 98.65]) and walking (M = 97.89% [97.42, 98.34]) ($p = .071$), nor between round (M = 98.62% [98.34, 98.90]) and square dials (M = 98.10% [97.74, 98.51]) ($p = .090$). However, visualization type had a significant effect on accuracy ($p = .002**$), with bar charts resulting in higher accuracy (M = 96.28% [95.80, 96.68]) than radial bar charts (M = 95.44% [94.96, 95.92]).

Task 2: Estimate Completed Percentage. **Completion Time:** Participants completed faster while walking (M = 1636.56ms [1557.97, 1705.23]) than when sitting (M = 1781.20ms [1691.34, 1863.79]); $p < .001***$, similar to Task 1. No significant differ-

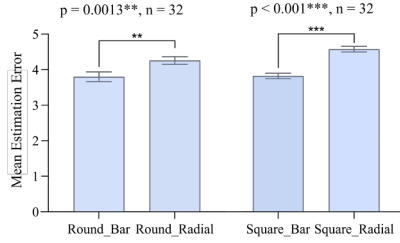


Figure 3: Mean Estimation Error (%) for Task 2 of the combination effect of Dial Shapes and Visualization Types. Error bars represent 95% Bootstrap confidence intervals (CIs) with p-value.

ence ($p = .21$) was observed between round $\odot$ ($M = 1581.51$ ms [1488.77, 1663.98]) and square dial $\square$ ($M = 1534.37$ ms [1446.61, 1610.64]). Additionally, visualization type significantly affected completion time ($p < .001^{***}$), with bar charts $\parallel$ enabling faster completion ($M = 1188.00$ ms [1117.73, 1249.20]) than radial bar charts $\odot$ ($M = 1555.53$ ms [1479.22, 1630.55]). *Estimation Error*: Participants exhibited significantly larger errors ($p < .001^{***}$), which mean while walking $\$$ ($M = 4.23\%$ [4.14, 4.33]) compared to sitting $\odot$ ($M = 3.83\%$ [3.75, 3.91]). A significant difference was also found between dials ($p = .003^{**}$), with round dials $\odot$ resulting in smaller errors ($M = 3.81\%$ [3.73, 3.90]) than square ones $\square$ ($M = 3.97\%$ [3.90, 4.04]). Similarly, visualization type had significant effect ($p < .001^{***}$), with radial bar charts $\odot$ yielding higher estimation errors ($M = 4.25\%$ [4.15, 4.35]) than bar charts $\parallel$ ($M = 3.57\%$ [3.51, 3.64]).

Univariate General Linear Model (GLM) Analysis on Task Performance. To further examine the combined effects of the three independent variables: motion, dial shape and visualization type, on user performance, we conducted a Univariate General Linear Model (GLM) analysis on Completion Time (ms) and Accuracy (%) in Task 1, Completion Time (ms) and Estimation Error (%) in Task 2. While most interactions between the three factors were not statistically significant, we found that there was a significant difference in Task 2 Estimation Error (see in Figure 3), between dial and visualization ($p = .012^*$). To explore this interaction, pairwise comparisons were conducted for each dial. Under the round dial $\odot$, participants performed significantly better using bar $\parallel$ charts ($M = 3.80\%$ [3.65, 3.95]) than radial bar charts $\odot$ ($M = 4.15\%$ [4.00, 4.29]) ($p = .0013^{**}$). Similarly, under the square dial $\square$, bar charts $\parallel$ ($M = 3.84\%$ [3.69, 3.98]) outperformed radial bar charts $\odot$ ($M = 4.56\%$ [4.41, 4.71]) ($p < .001^{***}$). These findings indicate that while radial bar charts $\odot$ generally resulted in lower error on the estimation task, this adverse effect was more pronounced when displayed on square dials, highlighting a notable dial design–visualization interaction effect.

Hypothesis Validation. We validated our hypotheses using the experimental results. **H1a** was supported: participants completed both tasks faster while walking. **H1b** was partially supported: motion reduced accuracy in Task 2 but not in Task 1. **H2** was not supported: round dials $\odot$ were better than square dials $\square$ on reading estimation in Task 2. **H3** was supported: bar charts $\parallel$ led to faster and more accurate performance in both tasks. **H4** was not supported: bar charts $\parallel$ on square $\square$ and round $\odot$ dials both outperformed radial bar charts $\odot$. **H5** was not supported: no evidence showed that radial bar charts $\odot$ on round dials $\odot$ outperformed than other combinations while walking $\$$.

Semi-Structure Interview Findings. For **RQ1**, all 32 participants (100%) reported that walking made it harder to focus on smartwatch visualizations, often describing the experience as “vision becomes slightly shaky while walking”. Additionally, 26 participants (81.3%) noted that they read faster during walking, stating that “when I am seated, I can look slowly and carefully; when I am walking, I can only glance quickly”. A further 24 participants (75.0%) explained that while seated, they “inspect each data

item one-by-one”, whereas under motion they relied on “a quick glance” strategy to capture key information. For **RQ2**, preferences for dial design were divided. Fourteen participants (43.8%) favored the square dial, appreciating its structured layout that “keeps the elements neatly aligned”. In contrast, 18 participants (56.2%) preferred the round dial, citing greater familiarity and aesthetic appeal, such as “round faces are more familiar for me” and “appear visually elegant and premium”. Regarding reading strategies, 22 participants (68.8%) on the square dial described using a “top-to-bottom, left-to-right comparison” approach, while 27 participants (84.4%) on the round dial reported “sweeping their gaze along the outline of the dial”. For **RQ3**, the majority of participants (25 out of 32, 78.1%) preferred bar charts, noting that “the tallest bar instantly reveals the maximum” and that “the design is more intuitive and straight”. The remaining seven participants (21.9%) favored radial bar charts, mainly for aesthetic reasons, although most acknowledged that “a full radial sweep requires more time for me”. In terms of estimation strategies, 31 participants (96.9%) reported mentally referencing the midpoint when interpreting bar charts, despite no explicit tick marks being shown. Similarly, 29 participants (90.6%) indicated that when reading radial bar charts, they internally aligned their estimations with imagined quadrant divisions (25%, 50%, and 75%) based on the chart’s overall structure.

5 DISCUSSION AND CONCLUSION

This study examined how motion, dial shape, and visualization type influence users’ interpretation of activity data on smartwatches. Our findings indicated that **motion** significantly affects reading performance: participants completed both tasks more quickly while walking, yet their accuracy—particularly in estimating percentages (Task 2)—declined, consistent with previous studies [15, 18]. While **dial shape** did not impact task completion time, it influenced accuracy in more complex task—Task 2 (reading specific values). Round dials led to slightly better reading accuracy. Notably, bar charts on square or round dials performed better than radial bar charts in Task 2. **Visualization type** did impact reading performance. Bar charts consistently outperformed radial bar charts in both speed and accuracy in two tasks, supporting prior findings on the perceptual benefits of linear layouts [10, 26, 5]. Participants also described bar charts as more intuitive for making quick estimates, aligning with established research that favors length-based over angle-based representations [8]. For future **design considerations**, we recommend using bar charts for goal-related tasks (such as checking and reading) on smartwatches. Furthermore, regardless of whether the dial is round or square, bar charts are a better choice than radial charts when precise value reading is important. When reading specific values during walking, purely visual presentations may not be sufficient; combining visualizations with speech feedback (e.g., voice-based reporting of key values) might further enhance performance.

This study has several limitations. The fixed treadmill speed may not fully capture the variability of real-world walking conditions, and our relatively young participant sample limits generalizability. While our design omitted interface interactions for simplicity, potentially limiting ecological validity, we note that in most practical contexts users primarily view data with little interaction. Our study therefore focuses on this typical usage pattern. Future work should explore diverse motion conditions, user groups, and long-term adaptation to visualization–dial combinations. Overall, our findings highlight how motion, dial shape, and visualization design jointly shape data interpretation on smartwatches, offering valuable guidance for creating more effective and adaptive visualizations in mobile contexts.

ACKNOWLEDGMENTS

We would like to thank Dr. Lijie Yao for her insightful suggestions regarding the experiment. The work was supported by XJTLU RDF (22-01-092).

REFERENCES

- [1] F. Bentley, K. Tollmar, P. Stephenson, L. Levy, B. Jones, S. Robertson, E. Price, R. Catrambone, and J. Wilson. Health mashups: Presenting statistical patterns between wellbeing data and context in natural language to promote behavior change. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 20(5):1–27. doi: 10.1145/2503823
- [2] T. Blascheck, L. Besançon, A. Bezerianos, B. Lee, and P. Isenberg. Glanceable visualization: Studies of data comparison performance on smartwatches. *IEEE transactions on visualization and computer graphics*, 25(1):630–640. doi: 10.1109/TVCG.2018.2865142
- [3] T. Blascheck, L. Besançon, A. Bezerianos, B. Lee, A. Islam, T. He, and P. Isenberg. Studies of part-to-whole glanceable visualizations on smartwatch faces. In *Proceedings of the IEEE Pacific Visualization Symposium*, pp. 187–196. IEEE Computer Society Press, Washington D.C., doi: 10.1109/PacificVis56936.2023.00028
- [4] T. Blascheck and P. Isenberg. A Replication Study on Glanceable Visualizations: Comparing Different Stimulus Sizes on a Laptop Computer. In *Proceedings of the Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications*, pp. 133–143. Science and Technology Publications, Virtual, Austria. doi: 10.5220/0010328501330143
- [5] M. Brehmer, B. Lee, P. Isenberg, and E. K. Choe. Visualizing ranges over time on mobile phones: a task-based crowdsourced evaluation. *IEEE transactions on visualization and computer graphics*, 25(1):619–629. doi: 10.1109/TVCG.2018.2865234
- [6] M. Burch, F. Bott, F. Beck, and S. Diehl. Cartesian vs. radial—a comparative evaluation of two visualization tools. In *Proceedings of Advances in Visual Computing: 4th International Symposium*, pp. 151–160. Springer. doi: 10.1007/978-3-540-89639-5_15
- [7] E. K. Choe, N. B. Lee, B. Lee, W. Pratt, and J. A. Kientz. Understanding quantified-selfers’ practices in collecting and exploring personal data. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, p. 1143–1152. ACM, New York. doi: 10.1145/2556288.2557372
- [8] W. S. Cleveland and R. McGill. An experiment in graphical perception. *International Journal of Man-Machine Studies*, 25(5):491–500. doi: 10.1016/S0020-7373(86)80019-0
- [9] P. Crowley, P. Madeleine, and N. Vuillerme. The effects of mobile phone use on walking: A dual task study. *BMC Research Notes*, 12:1–6. doi: 10.1186/s13104-019-4391-0
- [10] S. Diehl, F. Beck, and M. Burch. Uncovering strengths and weaknesses of radial visualizations—an empirical approach. *IEEE transactions on visualization and computer graphics*, 16(6):935–942. doi: 10.1109/TVCG.2010.209
- [11] D. Epstein, F. Cordeiro, E. Bales, J. Fogarty, and S. Munson. Taming data complexity in lifelogs: exploring visual cuts of personal informatics data. In *Proceedings of the Conference on Designing Interactive Systems*, p. 667–676. ACM, New York. doi: 10.1145/2598510.2598558
- [12] J. Goldberg and J. Helfman. Eye tracking for visualization evaluation: Reading values on linear versus radial graphs. *Information Visualization*, 10(3):182–195. doi: 10.1177/1473871611406623
- [13] R. Gouveia, E. Karapanos, and M. Hassenzahl. Activity tracking in vivo. In *Proceedings of the Conference on Human Factors in Computing Systems*, p. 1–13. ACM, New York. doi: 10.1145/3173574.3173936
- [14] F. Grioui and T. Blascheck. Study of heart rate visualizations on a virtual smartwatch. In *Proceedings of the ACM Symposium on Virtual Reality Software and Technology*. ACM, New York. doi: 10.1145/3489849.3489913
- [15] F. Grioui, T. Blascheck, L. Yao, and P. Isenberg. Micro visualizations on a smartwatch: Assessing reading performance while walking. In *IEEE Visualization Conference*, pp. 46–50. doi: 10.1109/VIS55277.2024.00017
- [16] A. Islam, R. Aravind, T. Blascheck, A. Bezerianos, and P. Isenberg. Preferences and effectiveness of sleep data visualizations for smartwatches and fitness bands. In *Proceedings of the Conference on Human Factors in Computing Systems*, pp. 1–17. ACM, New York. doi: 10.1145/3491102.3501921
- [17] A. Islam, T. He, A. Bezerianos, B. Lee, T. Blascheck, and P. Isenberg. Visualizing information on smartwatch faces: A review and design space. *arXiv preprint*. doi: 10.48550/arXiv.2310.16185
- [18] T. Mustonen, M. Olkkonen, and J. Hakkinen. Examining mobile phone text legibility while walking. In *Proceedings of CHI Extended Abstracts on Human Factors in Computing Systems*, p. 1243–1246. ACM, New York. doi: 10.1145/985921.986034
- [19] J. Niess, K. Knaving, A. Kolb, and P. W. Woźniak. Exploring fitness tracker visualisations to avoid rumination. In *Proceedings of the Conference on Human-Computer Interaction with Mobile Devices and Services*. ACM, New York. doi: 10.1145/3379503.3405662
- [20] B. Schildbach and E. Rukzio. Investigating selection and reading performance on a mobile phone while walking. In *Proceedings of the Conference on Human Computer Interaction with Mobile Devices and Services*, p. 93–102. ACM, New York. doi: 10.1145/1851600.1851619
- [21] M. Serrano, A. Roudaut, and P. Irani. Investigating text legibility on non-rectangular displays. In *Proceedings of the Conference on Human Factors in Computing Systems*, p. 498–508. ACM, New York. doi: 10.1145/2858036.2858057
- [22] M. Serrano, A. Roudaut, and P. Irani. Visual composition of graphical elements on non-rectangular displays. In *Proceedings of the Conference on Human Factors in Computing Systems*, p. 4405–4416. ACM, New York. doi: 10.1145/3025453.3025677
- [23] F. Simon, A. Roudaut, P. Irani, and M. Serrano. Finding information on non-rectangular interfaces. In *Proceedings of the Conference on Human Factors in Computing Systems*, p. 1–8. ACM, New York. doi: 10.1145/3290605.3300332
- [24] J. Talbot, V. Setlur, and A. Anand. Four experiments on the perception of bar charts. *IEEE transactions on visualization and computer graphics*, 20(12):2152–2160, 2014. doi: 10.1109/TVCG.2014.2346320
- [25] C. J. Turner, B. S. Chaparro, and J. He. Texting while walking: is it possible with a smartwatch? *J. Usability Studies*, 13(2):94–118.
- [26] M. Waldner, A. Diehl, D. Gračanin, R. Splechtna, C. Delrieux, and K. Matković. A comparison of radial and linear charts for visualizing daily patterns. *IEEE transactions on visualization and computer graphics*, 26(1):1033–1042. doi: 10.1109/TVCG.2019.2934784
- [27] Z. Xia, Y. Liu, and F. Polidoro. A study on activity visualization for smart watches. In *Proceedings of the IEEE Pacific Visualization Symposium*. doi: 10.48550/arXiv.2407.02012